

Attack Models for Digital Signatures

- Key only attack: adversary knows only the verification key.
- Known message attack: Adversary has a list of messages and corresponding signature.
 $(m_1, \sigma(m_1)), (m_2, \sigma(m_2)), \dots$
- Chosen message attack: adversary can request sig. on messages of choice.

Goals:

Total break: private key so that ~~any~~ signature can be forged over any

msg.

Selective forgery: able to create a valid signature on a message chosen by someone else with a significant probability.

Existential forgery: able to create valid sig. at least over one msg.

$$S = x^{e \cdot x^{e \bmod n}}$$

$(x, S) \longrightarrow \text{BOB}$

$$x^{ee \bmod n}$$

$$n = pq$$

$$\phi(n) = (p-1)(q-1)$$

$$\text{GCD}(e, \phi(n)) = 1$$

$$e \cdot d \equiv 1 \pmod{\phi(n)}$$

$$c_1 = m_1^e \pmod{n}$$

$$c_2 = m_2^e \pmod{n}$$

$$c_1 c_2 = (m_1 m_2)^e \pmod{n}$$

$$m_1, m_2$$

$$C = (m_1 m_2)^e \pmod{n}$$

✓

Ques RSA enary.

Show that multiplicative prop.
holds for RSA, i.e. show that the product
of two ciphertexts is equal to the encryption
of the product of two respective plaintexts.

Q Encrypt the following message with Elgamal
scheme ($p=467$ and $\alpha=2$)

$$(i) K_{p,r} = d = 105, \quad 29 \quad m = 33$$

ALICE

(α, β, p)

BOB

$$r = 213$$

$$\begin{aligned} Y_1 &= \alpha^k \pmod{p} \\ &= 2^{105} \pmod{467} \\ &= 92 \end{aligned}$$

$$\begin{aligned} Y_2 &= m \cdot \beta^k \pmod{p} \\ &= 33 \cdot (29)^{105} \pmod{467} \end{aligned}$$

$$\begin{aligned} \beta &= \alpha^{213} \pmod{p} \\ &= 2^{213} \pmod{467} \\ &= 29 \end{aligned}$$

Insecurity of plain RSA

- Forging a signature on arbitrary msg.

Two pairs of msg and signature
 $(m_1, s(m_1))$ and $(m_2, s(m_2))$

forges signature on m_3

$$m_3 = m_1 \cdot m_2 \bmod n$$

$$s(m_3) = s(m_1) s(m_2) \bmod n$$

$$m_1^d \bmod n \quad m_2^d \bmod n$$

$$(m_1 m_2)^d \bmod n$$

$$m_3^d \bmod n$$

ELGAMAL SIGNATURE SCHEME over $\mathbb{Z}_p$

ALICE
(set up phase)

$$e_A = (p, \alpha, \beta)$$

$$\boxed{\alpha_A = \alpha}$$

$$\boxed{\beta \equiv \alpha^a \pmod{p}}$$

$$p = \mathbb{Z}_p^*$$

$$A = \mathbb{Z}_p^* \times \mathbb{Z}_{p-1}$$

$x \in \mathbb{Z}_p^*$
message
or plaintext

Alice choose a random k from $\mathbb{Z}_{p-1}^*$

$$\text{sig}_{\alpha_A}(x, k) = (r, s) \checkmark$$

signature

$$\boxed{r = \alpha^k \pmod{p} \checkmark}$$

$$\boxed{s = (x - \alpha r) k^{-1} \pmod{p-1}}$$

$$\text{Ver}_{\alpha_A}(r, s) (x, \overline{r}, \overline{s})$$

$$\boxed{\beta r \overline{s} \equiv \alpha^x \pmod{p}}$$

BOB

choose p and α .
 $k_{pr} = a \in \{2, 3, \dots, p-2\}$
 $\beta = \alpha^a \pmod{p}$
 k_{pub}
 ephemeral key $k_E \in \{2, 3, \dots, p-2\}$
 s.t. $\text{GCD}(k_E, p-1) = 1$ $\downarrow$ msg specific

$$K = \{(p, \alpha, a, \beta)\}$$

$$\boxed{\beta \equiv \alpha^a \pmod{p}}$$

primitive element

sign msg using k

$$\beta^r = ((\alpha)^a)^r$$

∴

$$= (\alpha^{ar})$$

$$\gamma^s = (\alpha^{kr})$$

$$\alpha^{ar} \alpha^{kr} = \alpha^{ar+kr}$$

$$= \alpha^x$$

This one is easy
compare

$$kr = x - ar$$

$$kr + ar = x$$

$$r = (x - ar)k^{-1} \bmod p-1$$

multiply by k

$$kr = (x - ar) \bmod p-1$$

$$kr + ar = x \bmod p-1$$

Example

$$p = 467$$

$$x = 100$$

$$a = 2$$

$$a = 127$$

$$\beta = \alpha^a \bmod p$$

$$= 2^{127} \bmod 467$$

$$= 132$$

ALICE it has inverse

$$|k = 213$$

BOB

$$\gamma = 2^{213} \bmod 467 = 29$$

$$\delta = (100 - 127 \times 29)^{431} \bmod 466$$

$$= \cancel{431 \bmod 466} =$$

$$= 51$$

$$(x, \gamma, \delta)$$

$$\begin{matrix} 29 & 51 & 132 \\ 132 & 29 & = 2 \bmod 467 \end{matrix}$$

123¹³¹

proof of correctness

$$A_{LICE} \quad \beta^r \gamma^\delta = (\alpha^a)^r (\alpha^k)^\delta \text{ mod } p$$

$$\equiv \alpha^{ar} \alpha^{k\delta} \text{ mod } p$$

$$\equiv \alpha^{(ar+k\delta)} \text{ mod } p$$

$= \alpha^x$

[Fermat's little theorem

$$a^{p-1} \equiv 1 \text{ mod } p$$

$$\text{gcd}(a, p) = 1$$

any integer
(multiple
of $p-1$ + remainder)

$$a^m = a^{m \text{ mod } p-1} a^{m \text{ mod } p-1}$$

$$\equiv a^{2(p-1) + r}$$

$$\equiv (a^2)^{p-1} = a^2$$

$$\equiv (a^{p-1})^2 \cdot a^r$$

$$a^m \equiv 1 \cdot a^r \text{ mod } p$$

$$a^m \equiv a^{m \text{ mod } p-1} \text{ mod } p$$

on exponents we are allowed to

$$a^{r+k\delta} = x \text{ mod } p-1$$

$$k\delta \equiv x - ar \text{ mod } p-1$$

$$\delta = (x - ar)^{-1} \text{ mod } p-1$$

construction of

$$\delta$$

2	131	1
2	65	1
2	32	0
2	16	0
2	8	0
2	4	0
2	2	0

10000011

① $2 = 1$
 $2 = 123$
 ②

The signature
 (r, δ) has twice
 the bit length of
 x .

ex $p = 2^{2048}$
 $\Rightarrow \lceil (r + \delta) \rceil = 4096$ bits

* Elgamal digital
 signature is base for
 DSA $\rightarrow$ digital signature
 algo.

Weakness of (E, d, S) :- basis fees DSA

Reuse of k $\rightarrow$ let BOB uses k for two msg

$$x_1 \neq x_2$$

OSCAR $x_1, (y, \delta_1)$

$x_2, (\cancel{y}, \delta_2) (y, \delta_2)$

GOAL $\frac{x}{y} a = ?$

$$\delta_1 \equiv (x_1 - ay) k^{-1} \pmod{p-1}$$

$$\delta_2 \equiv (x_2 - ay) k^{-1} \pmod{p-1}$$

$$\delta_1 - \delta_2 \equiv (x_1 - x_2) k^{-1} \pmod{p-1}$$

$$k \equiv \frac{(x_1 - x_2)}{(\delta_1 - \delta_2)} \pmod{p-1}$$

once k is known

$$a \equiv \left(\frac{\delta_1 - x_1 k}{y} \right) \pmod{p-1}$$

Existential forgery attack

ALICE

BOB
 $k_{pub} (P, \alpha, \beta)$ $k_{pub} = (P, \alpha, \beta), k_{priv} = a$



OSCAR

Goal is to

generate valid pair

1. select u, v s.t. $\gcd(v, p-1) = 1$

2. $y \equiv \alpha^u \cdot \beta^v \pmod{p}$

$\delta \equiv -x y^{-1} \pmod{p-1}$

3. compute msg.

$x \equiv \delta u \pmod{p-1}$

(x, y, δ)

ALICE

$$\beta^r \gamma^s \bmod p-1$$

$$\beta \equiv \alpha^a \bmod p-1$$

$\Rightarrow$ valid

Validation

why does attack work?

$$\beta^r \gamma^s \equiv (\alpha^a)^r (\alpha^{u_0} \beta^j)^s$$

$$\equiv \alpha^{ar} \alpha^{u_0 s} \beta^{js}$$

$$\equiv \alpha^{\overbrace{(a+i)r} \quad \overbrace{j}s}$$

$$\equiv \alpha^{ar+js}$$

$$\equiv \alpha^{ar+js} (\alpha^a)^{js}$$

$$\equiv \alpha^{ar+js+a^j s}$$

$$\equiv \alpha^{ar} (\alpha^{u_0+a^j})^s$$

$$\equiv \alpha^{ar} \alpha^{(u_0+a^j)s}$$

$$\equiv \alpha^{ar} \alpha^{(u_0+a^j)(-rj^{-1}) \bmod p}$$

$$\equiv \alpha^{ar} \alpha^{-rj^{-1}u_0} \alpha^{-a^j r j^{-1} \bmod p}$$

$$\equiv \alpha^{\overbrace{-rj^{-1}u_0}}$$

$$\equiv \boxed{\alpha^{s_1} \bmod p}$$

$$\alpha^{s_1} \bmod p$$

but ALICE checks

$$\alpha = \alpha^{s_1}$$

$$p = 31$$

$$\alpha = 3, \beta = 6, x = 10$$

$$r = 13, s = 15$$

Verification

$$\beta^r \gamma^s \equiv \alpha^x \pmod{p}$$

$$6^{13} 13^{15} \equiv 3^{10} \pmod{31}$$

2	10	0
2	5	1
2	2	0
	1	

$$1010$$

$$(1) z = 1$$

$$z = 2$$

$$(2) z = 9$$

$$z = 9$$

$$(3) z = 81 \pmod{31}$$

$$= 19$$

$$19 \times 3 \pmod{31}$$

$$57 \pmod{31}$$

$$26$$

$$26 \times 26 \pmod{31}$$

$$676 \pmod{31}$$

$$25$$

$$6^{13}$$

$$13$$

$$1101$$

$$(1) 6$$

$$(2) 5$$

$$30$$

$$(3) 900 \pmod{31}$$

$$1$$

$$(4) z = 1$$

$$z = 6$$

$$15 = 1111$$

$$(1) z = 1$$

$$z = 13$$

$$(2) z = 169 \pmod{31}$$

$$= 14$$

$$z = 14 \times 13 \pmod{31}$$

$$= 27$$

$$(3) z = 16$$

$$z = 16 \times 13 \pmod{31}$$

$$= 22$$

$$(4) z = 22 \times 22 \pmod{31}$$

$$= 44$$

$$44$$

$$484 \pmod{31}$$

$$= 19$$

$$19 \times 13 \pmod{31}$$

$$57$$

$$19$$

$$247 \pmod{31}$$

$$30$$

$$P = 31, \alpha = 3, \beta = 6, x = 10$$

$$Y = 17, \delta = 5$$

$$\beta^Y \gamma^\delta \equiv \alpha^x \pmod{P}$$

$$6^{17} 17^5 \equiv \boxed{3}^{10} \pmod{P}$$

$$6^{17} 17^5 \pmod{31} \equiv 3^{10} \pmod{31}$$

2	17	1
2	8	0
2	4	0
2	2	0
	1	

$$36 \times 9 \pmod{31}$$

$$324 \pmod{31} \equiv 25^x$$

$$\boxed{14}$$

$$10001$$

$$\textcircled{1} \quad z = 1$$

$$z = 6$$

$$\textcircled{2} \quad z = 36 \pmod{31}$$

$$5 \checkmark$$

$$2 \times 5 \times 6 = 30$$

$$\textcircled{3} \quad 25 \checkmark$$

$$\textcircled{4} \quad 625 \pmod{31}$$

$$5 \checkmark$$

$$\textcircled{5} \quad 3 \times 25 \pmod{31}$$

$$150 \pmod{31}$$

$$26$$

$$\frac{156}{124} = 6$$

$$\begin{array}{r} 2 \overline{) 100} \\ 2 \overline{) 51} \\ 2 \overline{) 20} \\ 1 \overline{) 10} \\ 1010 \end{array}$$

$$\textcircled{2} \quad \boxed{3}$$

$$\textcircled{12} \quad 9$$

$$\textcircled{3} \quad 81 \pmod{31}$$

$$\begin{array}{r} 54 \\ 31 \overline{) 18} \\ 13 \end{array}$$

$$18 \pmod{31}$$

$$54 \pmod{31}$$

$$23$$

$$\textcircled{4} \quad \boxed{95} \pmod{31}$$

$$\boxed{2}$$

$$17^5$$

$$0101$$

$$\textcircled{1} \quad z = 1$$

$$\textcircled{ii} \quad 17 \pmod{31}$$

$$17$$

$$\textcircled{3} \quad 289 \pmod{31}$$

$$779$$

$$\boxed{10}$$

$$\textcircled{4} \quad 100 \pmod{31}$$

$$7$$

$$17 \times 6$$

$$102 \pmod{31}$$

$$\boxed{9}$$